

STM32F103C8T6 Microcontroller



Indian Institute of Technology
Hyderabad

PCB Assignment : 02

EE4130: PCB Design

Harshil Rathan Y

EE24BTECH11064

भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

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1 Introduction

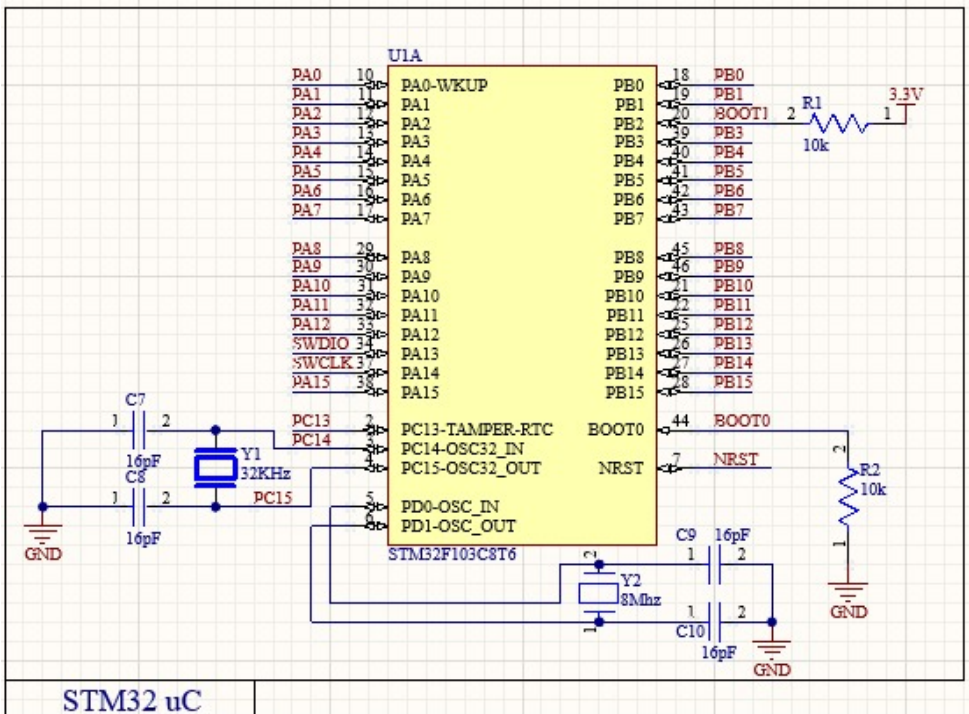
The STM32F103C8T6 microcontroller is a widely used 32-bit ARM Cortex M3 device suitable for embedded control, signal processing, and general purpose applications. This assignment focuses on designing a complete microcontroller PCB that adheres strictly to the manufacturer's datasheet and reference manual recommendations.

2 Objectives

The objectives of this assignment are

- To design a reliable 3.3 V power supply derived from a 5 V DC source.
- To correctly implement STM32 power pins, decoupling capacitors, reset circuitry, and clock sources.
- To provide a Serial Wire Debug (SWD) interface for programming and debugging.
- To prepare the design for fabrication on a two-layer PCB.
- To design at least one custom footprint. (USB Interface)

3 STM32 Micro-controller Overview



3.1 Important Datasheet Considerations

The following key points were considered during the schematic

Operating Voltage Requirements

It operates within a supply voltage range of 3.0 V to 3.6 V, with 3.3 V as the nominal operating voltage. Direct connection to a 5 V supply is not permitted, therefore, an external voltage regulator is required to generate a stable 3.3 V supply from the 5 V input.

Power Pin Connections

The microcontroller contains multiple VDD and VSS pins, all of which must be connected to ensure proper internal operation. The VDDA pin supplies the analog circuitry and is connected to the 3.3 V rail, while VSSA is connected to ground. Proper connection of these pins is essential for stable system behavior and accurate analog performance.

Decoupling Capacitors

According to the datasheet, each VDD pin must be decoupled using a 100 nF ceramic capacitor placed as close as possible to the corresponding pin. Additionally, a bulk capacitor is placed on the 3.3 V supply rail to handle transient current demands. Proper decoupling minimizes noise, voltage dips, and unintended resets.

VBAT Pin Handling

The VBAT pin supplies the backup domain, including the real-time clock and backup registers. In the absence of an external battery, VBAT is connected to the 3.3 V supply and decoupled using a 100 nF capacitor to ground to ensure stable operation of the backup domain.

Reset Circuitry

The NRST pin is an active-low reset input that must be pulled up to the supply voltage. A 10 k Ω pull-up resistor and a 100 nF capacitor to ground are used to provide reliable reset behavior and noise filtering. A manual reset push button is also included for user-initiated resets.

Boot Configuration Pins

Boot mode selection is controlled using the BOOT0 pin and the BOOT1 function on pin PB2. To ensure normal execution from internal Flash memory, BOOT0 is pulled down using a 10 k Ω resistor. BOOT1 is also pulled down to guarantee deterministic boot behavior. Floating boot pins are avoided to prevent unintended boot modes.

Clock Source Selection

The STM32F103C8T6 supports both internal and external clock sources. An external high-speed crystal oscillator is used to provide an accurate system clock, along with appropriate load capacitors as specified by the crystal manufacturer. Proper clock design ensures stable timing and reliable peripheral operation.

Programming and Debug Interface

The microcontroller supports Serial Wire Debug (SWD) for in-system programming and debugging. The SWD interface uses PA13 as SWDIO and PA14 as SWCLK. These signals, along with 3.3 V and ground, are routed to an external header to enable programming using an external debugger.

GPIO Pin Handling

Unused GPIO pins are intentionally left unconnected in the hardware design and are configured as analog inputs in firmware to reduce power consumption and prevent floating digital inputs. Required GPIO pins are either connected to peripheral headers or assigned to specific onboard functions.

- All the connections strictly adhere to the datasheet instructions.

4 Design Schematic

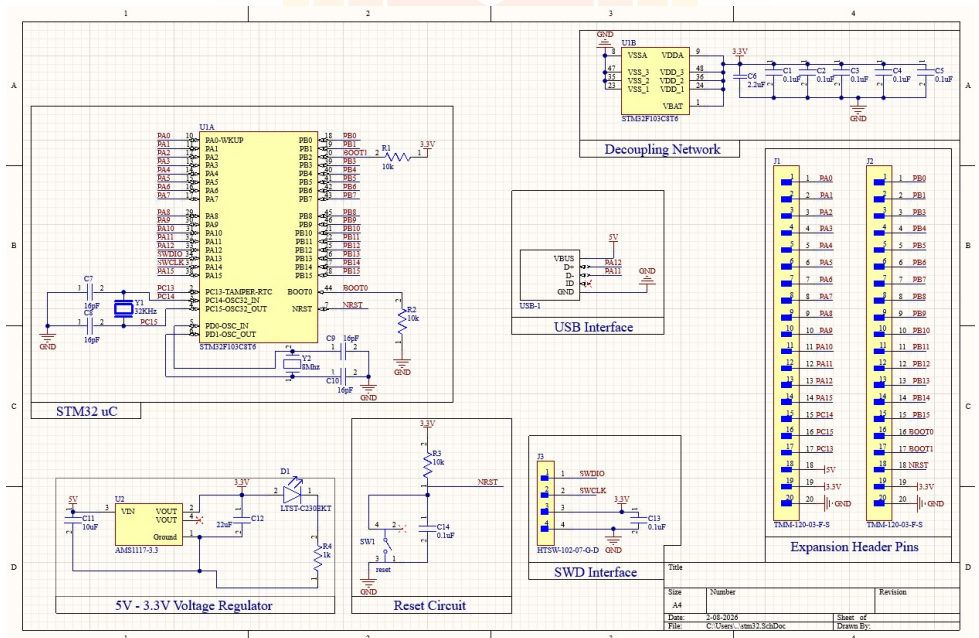
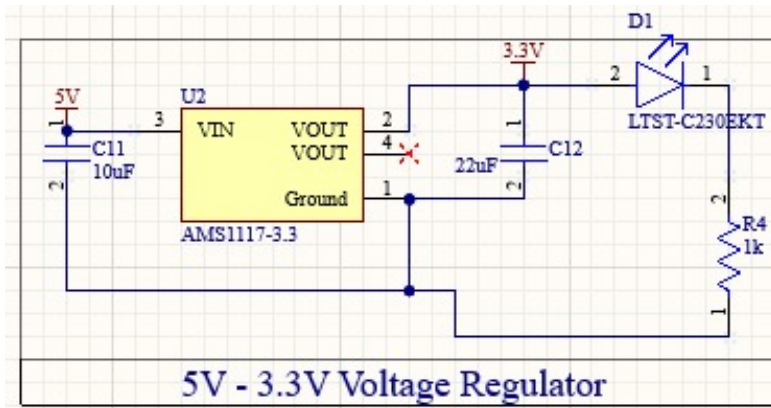


Fig. 0.1: Full Circuit Schematic

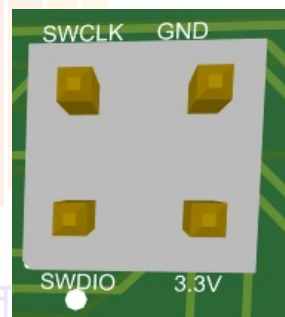
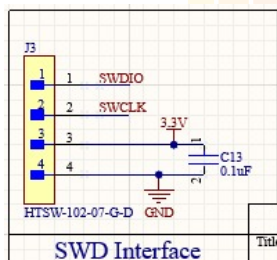
4.1 5V - 3.3V Voltage Regulator



From the datasheet, for a fixed output regulator for AMS1117 the following bypass capacitors are recommended

- Input Bypass Capacitor - **10uF**
- Output Bypass Capacitor - **22uF**
- A RED led is also added at the ouptput of the AMS1117 to indicate that the circuit is powered up.
- 1K Resistor is a pull up resistor for the red led.

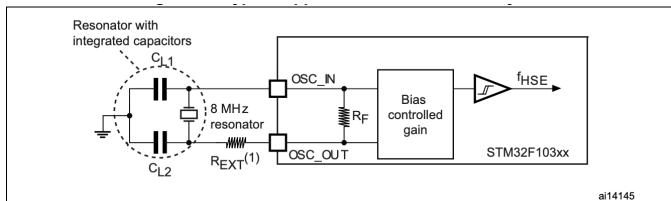
4.2 SWD Interface



- The SWD interface is a two-wire programming and debugging interface supported by the STM32 microcontroller.
- The SWD interface uses fixed pins, where PA13 functions as SWDIO (data line) and PA14 functions as SWCLK (clock line), as specified in the datasheet.

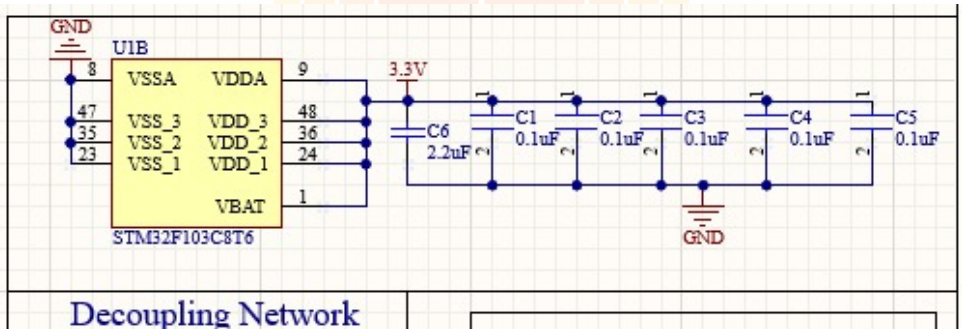
- As per the instructions a dedicated SWD header is provided on the PCB exposing SWDIO, SWCLK, 3.3 V, and GND signals to allow connection to an external debugger
- The 0.1uF capacitor helps suppress noise and voltage fluctuations that may be introduced.

4.3 8MHz Crystal Oscillator



- The datasheet recommends adding Capacitors of the range 5 pF to 25 pF
- 16pF Capacitors are added to meet the requirements.

4.4 Power Supply and Decoupling Capacitors



- To maintain a steady 3.3V voltage supply necessary decoupling capacitors 0.1uF are added.
- Recommended by the datasheet to suppress high-frequency noise on the power supply
- An additional bulk capacitor 2.2uF is included on the 3.3V rail to stabilize the supply during lower-frequency load variations
- Proper decoupling ensures reliable clock operation and overall stability

4.5 Expansion Headers

P1	P2
1 - SW0	1 - PB0
2 - SW1	2 - PB1
3 - SW2	3 - PB2
4 - SW3	4 - PB3
5 - SW4	5 - PB4
6 - SW5	6 - PB5
7 - SW6	7 - PB6
8 - SW7	8 - PB7
9 - SW8	9 - PB8
10 - SW9	10 - PB9
11 - SW10	11 - PB10
12 - SW11	12 - PB11
13 - SW12	13 - PB12
14 - SW13	14 - PB13
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95 - SW94	95 - PB94
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99 - SW98	99 - PB98
100 - SW99	100 - PB99

Expansion Header Pins

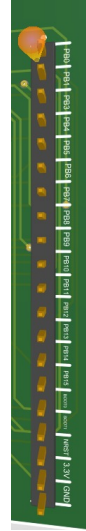
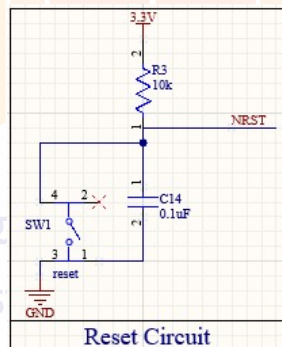


Fig. 0.2: SWD and GPIO expansion headers implemented in the PCB design

- Expansion headers are provided to expose unused GPIO pins of the microcontroller for external interfacing and future system expansion
- These can be easily connected to sensors, displays, communication modules, and user input devices without modifying the PCB.

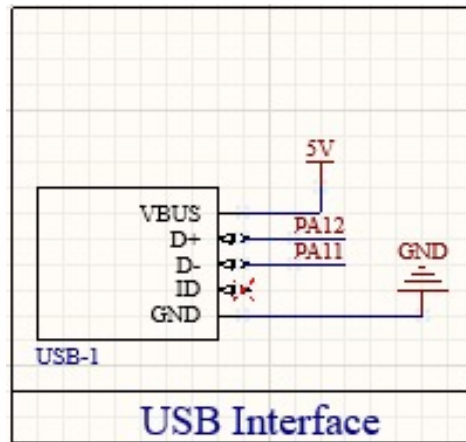
4.6 Reset Circuit



- A manual push-button reset is included to allow the user to intentionally reset the system during debugging or testing.

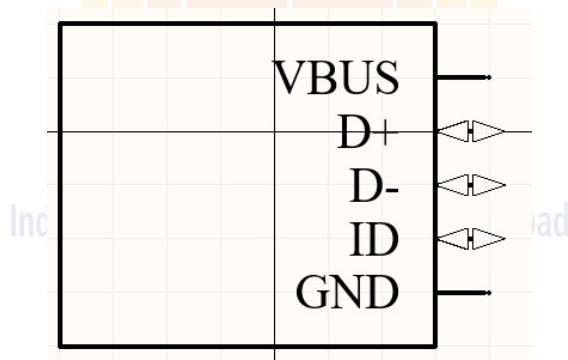
- A 10 k Ω pull-up resistor is connected to the NRST pin to ensure that the uC remains in the non-reset state during normal operation and to prevent the reset pin from floating.
- A 0.1 μ F capacitor is connected between the NRST pin and ground to filter noise and suppress anonymous resets.

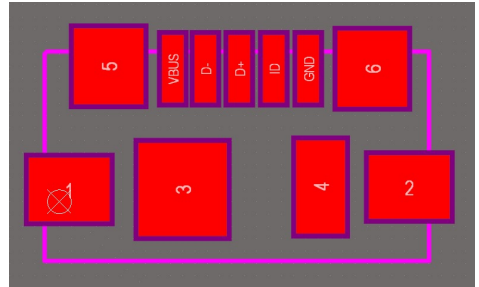
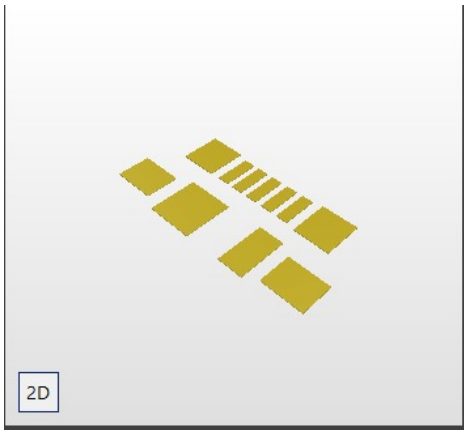
4.7 USB Interface



- The board is powered through a Micro-USB connector, which supplies a regulated 5 V input.
- AMS117 is used to supply a steady 3.3V which is the operating voltage of the STM32 Micro controller

5 Custom Footprint Design - USB Interface



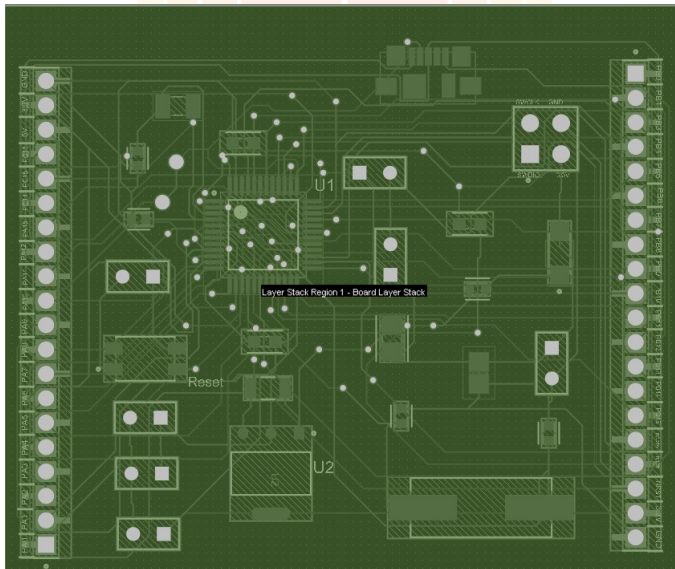


USB Connector PCB Placement

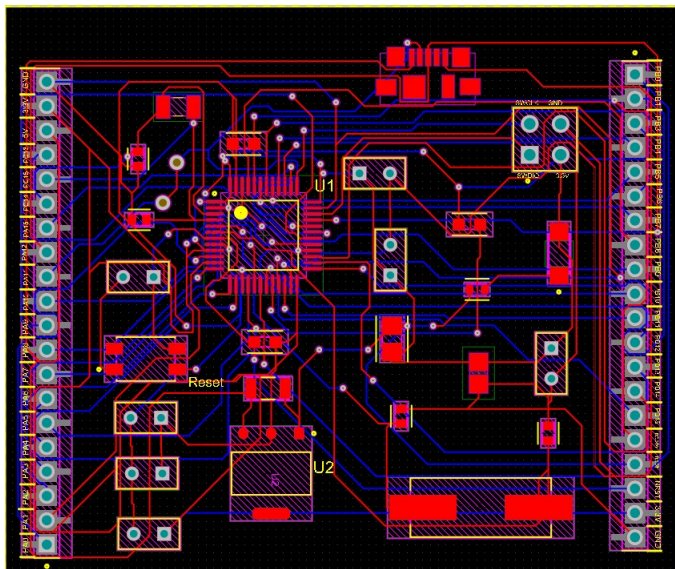
- A custom footprint of a USB connector was designed to satisfy the assignment requirement.
- The footprint was developed using the mechanical dimensions and pin specifications provided in the connector datasheet to ensure accurate pad placement.
- Proper pad sizing and silkscreen markings were included according to the datasheet.
- The footprint was verified by placing it on the PCB layout and confirming correct alignment with the schematic symbol.

6 PCB Layout

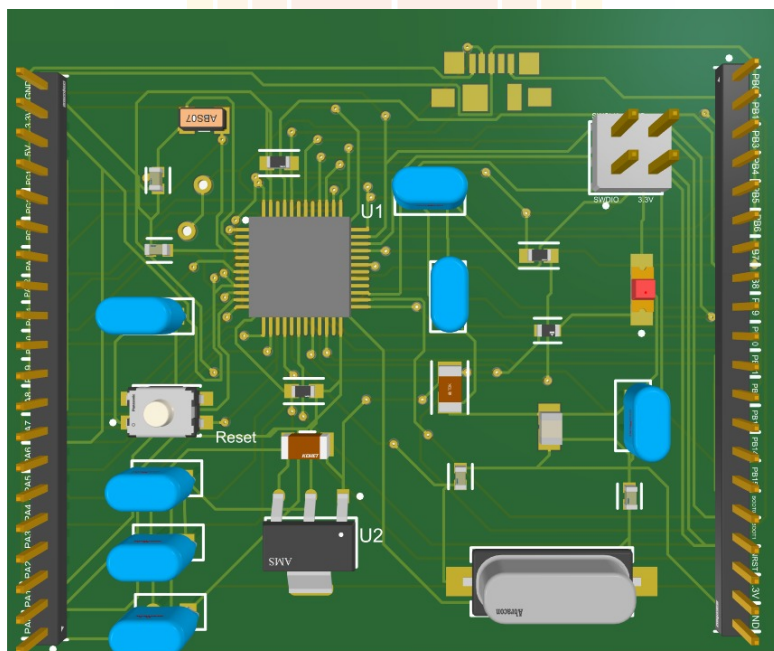
6.1 1D Layout

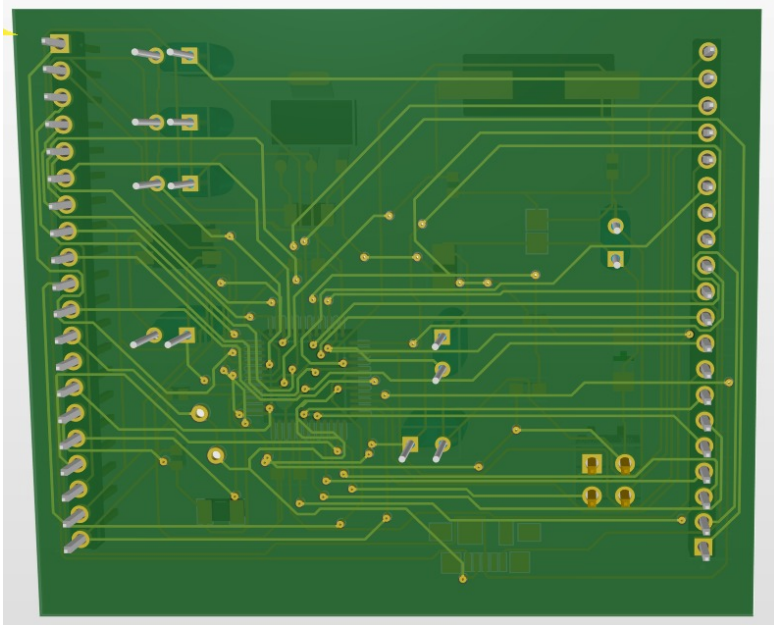


6.2 2D Layout



6.3 3D Layout





7 References

- USB Custom Footprint Datasheet
- STM32 Datasheet
- STM32 Youtube Walkthrough
- Custom Footprint - YT Tutorial

8 Conclusion

A complete two-layer PCB design based on the STM32 microcontroller was successfully implemented strictly following the datasheet. The design includes a regulated 3.3 V power supply derived from a 5 V input, proper power pin connections with decoupling capacitors, reset and boot configuration circuitry, and an external clock source.

A Serial Wire Debug (SWD) interface exposing SWDIO, SWCLK, 3.3 V, and ground was provided for programming and debugging. Expansion headers were added for GPIO accessibility, and a custom USB connector footprint was designed in Altium Designer to satisfy the assignment requirements.

A 2 layer PCB was successfully designed without violating any design rule checks, 0 design rule errors.

- The Gerber files were successfully generated verified, compressed in zip .
- Overall, the project results in a reliable, and manufacturable STM32 uC PCB